Answer **all** questions

Question 1

(a) Restriction fragment length polymorphism, RFLP, refers to the differences in restriction fragment length generated by restriction enzymes. Explain two ways in which the approach employed during RFLP analysis for the process of disease detection and for DNA fingerprinting differ. [4]

	Disease detection	**DNA fingerprinting**
Number of DNA probes used	One probe;	Multiple/ 10 or more probes;
➔Reason	Only one loci is analysed in disease detection;	multiple loci are analysed in the creation of a DNA fingerprint;
Location where restriction enzymes cut	Restriction enzymes cut within the gene locus	Restriction enzymes cut outside the Variable number of tandem repeat (VNTR) loci
➔ Reason	Changes in base sequence changes restriction sites that result in different sized restriction fragments being released when cut with the restriction enzyme.	The DNA polymorphism for this loci arises due to different number of tandem repeats of a particular sequence. Cutting outside the VNTR loci releases fragments of different lengths for each allele. Gel electrophoresis can then separate the different alleles by size.

(b) An autosomal recessive disease is caused by a mutation that causes a loss of a *BamH1* restriction within the gene locus as shown in Fig 1.1. The diagram shows the two different alleles of the gene being detected by a gene probe. (not drawn to scale)

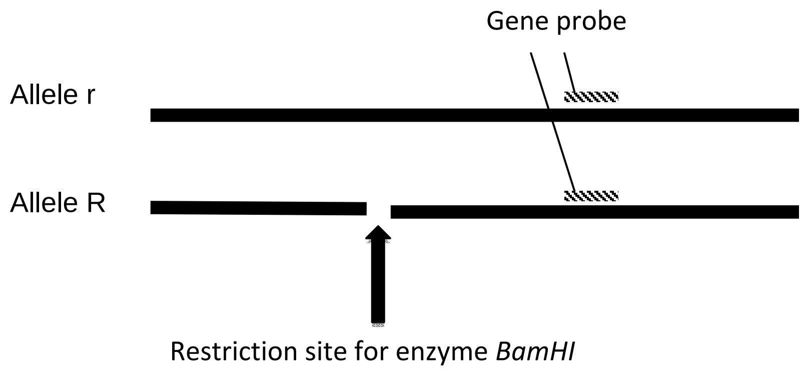


Fig. 1.1

(i) Explain how this gene probe can be used to detect which individuals are carriers for the disease. [3]

- For allele R, cutting the loci with *BamHI* will give rise to 2 smaller restriction fragments;
- While for allele r, one longer restriction fragment will be formed;
- After gel electrophoresis and nucleic acid hybridisation using the gene probe, homozygotes RR and rr will give one band;
- Given that the trait is recessive, carriers for the disease will be heterozygotes;
- Carriers will give 2 bands on the autoradiograph;
- And they will give one longer band due to presence of one r allele;
- And one shorter band due to the presence of one R allele;

(c) A particular region where there is variable number of tandem repeats of a CGG trinucleotide has been proposed to be a good RFLP marker to detect another disease where the gene sequence is unknown. The unique sequence outside the repeat region is as follows:

5' CAGTATGCA------(CGG)n-------ATGCGTAAT 3'
3' GTCATACGT------(GCC)n-------TACGCATTA 5’

(i) Suggest two factors that are required in order for a RFLP marker to be useful in detection of the disease. [2]

- The marker must be polymorphic, meaning that alternative forms must exist among individuals for detection among different members in the family.
- The marker must be closely linked to the disease locus such that crossing over between the marker and the gene is unlikely to occur during gamete formation. The greater the distance between the RFLP and gene locus, the lower the probability of an accurate diagnosis of the disease based on the RFLP marker.

(ii) Explain why it is necessary to know the sequence outside the repeat region in order to use it during RFLP analysis for the process of disease detection. [3]

- The different alleles of the region differ in the number of repeats of CGG;
- The sequence outside the repeat region is needed in order to determine an appropriate restriction enzyme that can cut there;
- to release the alleles/ entire repeat lengths as a unit;
- so that the different alleles can be separated and identified
- according to size during gel electrophoresis;
- by analysing which alleles of this RFLP marker individuals have, we can determine if the individuals are diseased.

(d) Shown in Fig 1.2 is the outline of a pedigree for a certain X-linked dominant disease and a representation of a gel showing PCR-amplified fragments detecting the number of AAGGTT repeats in the RFLP marker used as a proxy for disease detection. Repeats of more than 42 at the RFLP marker region are associated with the disease. The DNA corresponding to each individual is directly below his or her place in the pedigree.

(i) Based on the information you have been given, fill in the pedigree to show the sex of the individuals I-1, I-2, II-3, II-4, as well as their phenotype (affected vs. unaffected). [2]

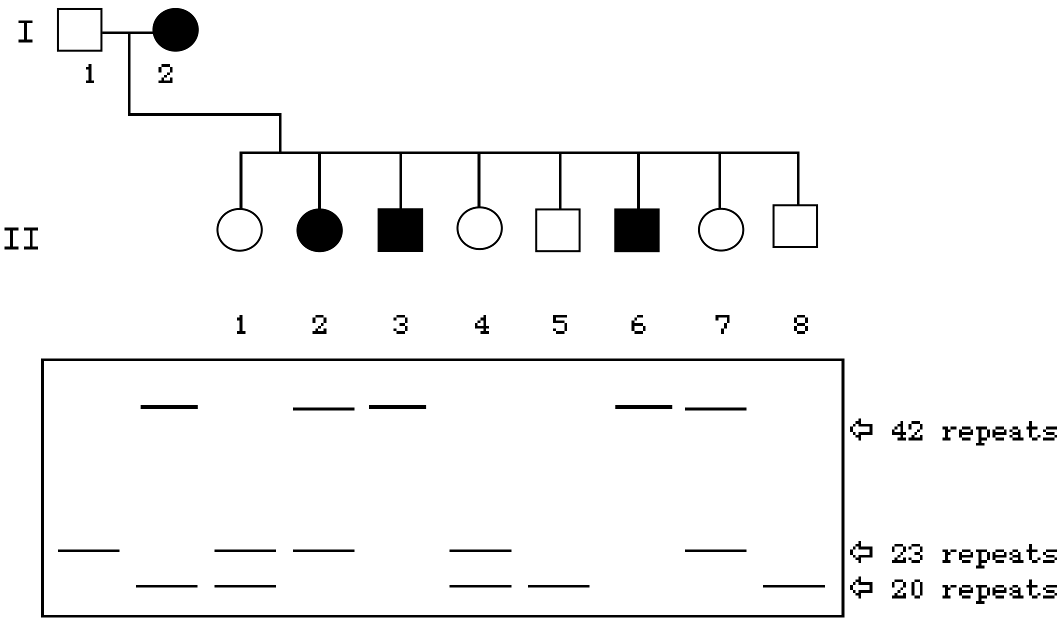


Fig. 1.2

(ii) In which individual did crossing over take place between the RFLP marker and the gene loci? [1] Answer: **I-2**

(iii) Explain your answer. [3]

- The diseased dominant allele is linked to the RFLP allele with more than 42 repeats;
- the father I-1 is hemizygous so crossing over between the non-sister chromatids of the X- chromosomes can only take place in the mother I-2;
- Individual I-2 is a heterozygote with one diseased allele and one recessive allele linked to the RFLP allele with 20 repeats;
- Crossing over occurs in meiosis 1 during the formation of gametes;
- The chance event of crossing over resulted in the formation of a recombinant gamete;
- Where the RFLP allele of more than 42 repeats is no longer linked to the diseased allele/ the RFLP allele of more than 42 repeats is now in a new linkage group with the non-diseased allele;
- Which was passed to daughter II-7 who does not have the disease but has inherited the RFLP allele of more than 42 repeats.

[Total: 17]

Question 2

Severe combined immunodeficiency (SCID) is a genetic disorder characterised by the absence of functional T-lymphocytes, resulting in the impairment of the immune system. There are different forms of SCID, each caused by mutation to a different gene. Two examples are X-linked SCID and ADA-SCID.

(a) Compare the differences between the two forms of SCID mentioned in the paragraph above. [4]

	X-linked SCID	**ADA-SCID**
Gene involved	• Gene coding for **interleukin 2 receptor gamma** [1/2]	• Gene coding for **adenosine deaminase** [1/2]
Chromosome on which the gene is found	• **X chromosome** [1/2]	• **Chromosome 20** [1/2]
Effect of mutation leading to SCID	• Results in production of **defective interleukin receptor** [1/2] • Leading to **defective signalling pathway** which **prevents proper development** of T-lymphocytes [1/2]	• Mutated enzyme results in **defective purine metabolism** [1/2] • Leading to the **accumulation of deoxyadenosine** which is **toxic** to immature lymphoid cell [1/2]

(b) ADA-SCID can be treated using *Ex vivo* gene therapy. T-lymphocytes from the patient are isolated from the patient's blood. Retroviruses are then used to introduce the normal allele into the T-lymphocytes to allow normal development. These T-lymphocytes are then introduced back into the patient's blood. This process needs to be repeated every few months in order to prevent the re-occurrence of the disease.

(i) Based on your knowledge of the structure of a retrovirus, explain why it serves as an efficient vector of gene therapy. [2]

- It contains **glycoprotein** that is **specific** to the cell-surface receptors on **T-lymphocytes** [1/2]
- This increases the **efficiency** of gene delivery to **specific target cell** [1/2]
- It contains the enzyme **integrase** [1/2]
- Which aids in the **integration** of the normal allele into the **genome of the T-lymphocyte** [1/2]

(ii) Suggest an advantage in carrying out *Ex vivo* gene therapy in the treatment of ADA-SCID. [1]

- **No immune response** against viral vector [1/2] as it is introduced to the T-lymphocytes **outside the body** [1/2]

- **Higher efficiency** in delivering the normal allele to T-lymphocytes [1/2] as the T-lymphocytes are **in higher number** than in blood [1/2]
- T-lymphocytes in which the normal allele has been successfully inserted into the host genome can be **selected and cultured** [1/2] to **increase the number of treated cells** being re-introduced into the patient's body [1/2].

(iii) Besides the need for repeated treatment, explain the other factors that keep this method of gene therapy from becoming an effective treatment for ADA-SCID. [2]

- **Low number of T-lymphocytes with the normal allele** compared to the high number of diseased cells at a tissue-level [1/2] so the effect of gene therapy is **low at a tissue-level** [1/2]
- **Difficult to control the expression** of the normal allele in T-lymphocytes [1/2] so they **may not synthesise sufficient functional proteins** to bring about an improvement [1/2]
- Integration of the normal allele into the genome of the T-lymphocytes may cause **cancer** [1/2] by **activating oncogenes or insertional inactivation of tumour suppressor genes** [1/2]

(c) In order to reduce the need for repeated treatment, a new method of gene therapy has been developed, in which the normal allele is introduced into blood stem cells instead of T-lymphocytes.

Distinguish between a blood stem cell and a T-lymphocyte. [3]

	Blood stem cell	**T-lymphocyte**
Where it is found	• **Bone marrow / umbilical cord blood** [1/2]	• **Blood** [1/2]
Differentiation	• **Undifferentiated** cell which is **mutlipotent** [1/2] • Able to differentiate into a **limited range of cell types** [1/2] • **Does not have tissue-specific structures** [1/2] (Max 1/2)	• **Differentiated** cell which is **unable to differentiate** [1/2] • Have **tissue-specific structures** [1/2] (Max 1/2)
Ability to divide	• Able to undergo mitotic cell division for **self-renewal** [1/2] • Have **telomerase activity** to allow many cell divisions [1/2] (Max 1/2)	• **Limited ability to divide** [1/2] • Have **limited / no telomerase activity** [1/2] (Max 1/2)
Function	• Differentiate into different types of blood cells for **replacement of damaged / worn-out blood cells** [1/2]	• Trigger **immune response** [1/2]

[Total: 12]

Question 3

Growth hormone genes from human, bovine or salmonid are introduced to farmed Atlantic salmon eggs to form transgenic fish that results in 10-80% increased growth rate than non-transgenic fish. Furthermore, fusing the gene with a strong gene promoter (e.g. ocean pout antifreeze promoter) further increases growth by 5 to 30 folds.

(a) State one method of introducing the recombinant vector containing the transgene into the fish egg. [1] Answer: Micro-injection; electroporation

Feedback / comment
Note that gene gun was originally developed for introducing foreign DNA into plant cells. Also be mindful of the difference between "electroporation" vs "electrophoresis"

The company behind the genetically modified (GM) Atlantic salmon believes that it now stands on the verge of an historic decision by the powerful US Food and Drug Administration (FDA) that will open the way to the sale of genetically engineered meat and fish both in the United States and the rest of the world.

b) (i) Explain how the approval by FDA may benefit the world.

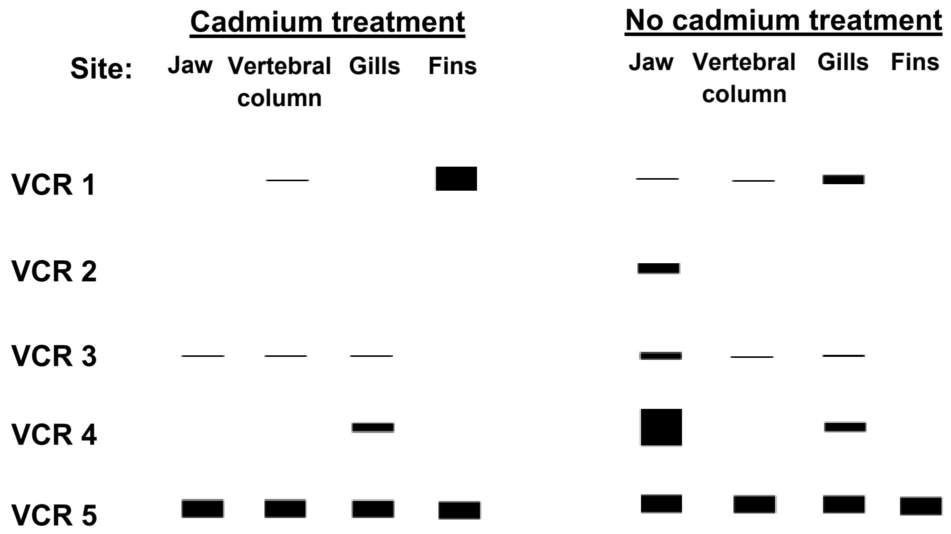
Since the GM fish is able to grow / mature faster, it will **increase salmon production** / **yield** [1/2] → help **reduce global food shortage** / **meet growing food demands** / **reduces** the **production cost of the fish** / the price of the fish will be kept low / affordable [1/2]

(ii) With reference to Fig 3.1, suggest a likely ethical reason for the protest. [1]
Suffering of the fish due to its **spinal deformities** which might affect its **ability to swim properly** [1]

R: Harming the animal

c) In response to the protest, the company issued a public statement that the deformities are not due to the genetic modifications but on the presence of cadmium ions in the local water supply. These ions affected the expression of a group of regulatory genes (VCR1, 2, 3, 4 and 5), known to be involved in the development of the fish.

The level of mRNA of the different genes in the farmed GM salmons with and without treatment of cadmium ions was measured in specific regions of the fish body.



(i) With reference to Fig 3.2, describe the effect of cadmium on the expression of the VCR genes for jaw development.

Cadmium ions **switched off expression** of **VCR1, 2 and 4 genes** completely (reference to absence of bands) [1/2]
Lowered expression of VCR 3 gene (reference to thinner band) [1/2]

No effect on VCR 5 gene expression (reference to same thickness of band) [1/2]
QV: reference to presence/absence or thickness of relevant bands [1/2]

(ii) Base on Fig 3.2 and the information provided, explain if the spinal deformities in the GM fish is likely to be due to the effect of cadmium. [2]

No / not likely [1/2]
Cadmium ions **did not affect the expression of all 5 VCR genes** [1/2] at the **vertebral column** [1/2]
mRNA band pattern remained the **same** when compared to no cadmium ions treatment [1/2]

(Also accept: VCR 1, 3 and 5 genes expressed at same level [1/2] while VCR gene 2 and 4 remained unexpressed [1/2] at vertebral column [1/2])

(iii) A cDNA library for all the VCR genes was created for further studies. Explain which specific part of the fish was initially used as the starting biological sample. [2]

Jaw [1/2]
It is because **all 5 genes** are **expressed** / transcribed [1/2]
Mature mRNA is the **template** to make the **complementary DNA (cDNA)** [1/2] via **reverse transcriptase** / through the process of **reverse transcription** [1/2]

d) The company claimed that all its GM salmons are genetically engineered to be sterile. Explain why this will reduce the social implication of the GM organism. [2]

For the GM salmon, the **rapid growth of the fish** suggests a competitive **advantage over wild fish** should the GM fish **escape** / The GM fish **may displace or prey upon smaller, slower growing individuals of the same species or other species** [1/2]

Since the **GM fish are sterile**, they would **not be able to interbreed** [1/2] with members from the wild or with each other and **pass the transgenes to** the **next generation** [1/2].

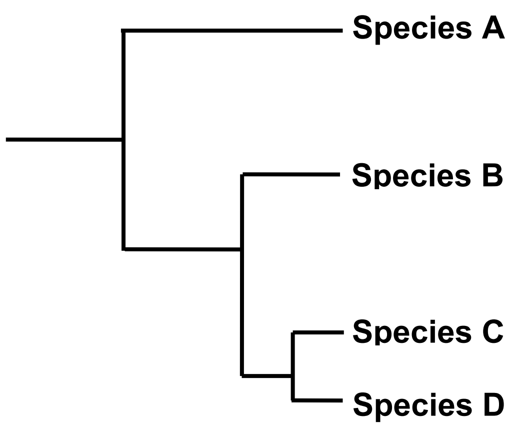
Hence, their **impact on the ecological balance** would be **minimal** / **reduced** [1/2]

[Total: 11]

Planning Question

4. The Orchidaceae, known as the orchid family, underwent a spectacularly diverse radiation since its late Cretaceous origin 83-75 million years ago to more than 27000 different species today. Classifying orchid species based on their evolutionary relationship is difficult and dynamic because of the large diversity of orchid species.

The phylogenetic tree of the four known Orchid species (A, B, C and D) is shown.



Using this information and your own knowledge, design an experiment to show the molecular homology between four species of orchids.

[Total 12]

Answer:

The evolutionary relationship between different orchid species can be determined by comparing the **similarity of DNA sequence of a common / conserved gene** (or non-coding region of the genome). **Closely related orchid species** will have very **similar DNA sequence** whereas **distantly related species will have more different DNA sequences.** **[1]**

Gel electrophoresis may distinguish distantly related species if the difference in DNA sequences results in **distinct sized DNA bands** after PCR. This could be due to **deletion or duplication of specific section of the gene**. Alternatively, the **differences in base sequences** may result in the **formation or deletion of a restriction site** which will result in **different restriction fragments after digestion** with the appropriate restriction enzyme HindIII. **[1]**

Procedure:

Extraction of DNA [1]
1) **10g** of orchid tissue samples from the 3 species were **homogenized** separately using **pestle and mortar**.
2) Add **10ml DNA extraction buffer solution** to the homogenized tissue. Stir the mixture using glass rod.

Preparation of DNA [1]
3) Add **1ml of the mixture** is added to a centrifuge tube and the mixture is **centrifuged**.
4) The **supernatant containing DNA** is then transferred to a fresh centrifuge tube.

Amplification of DNA
5) Use a thermal cycler to amplify the common gene of the 3 species using **appropriate set of DNA primers flanking the common DNA sequence.**

Note: PCR - Do not forget to mention "per cycle" and "total no. of cycles"

PCR step / *cycle*	**Temperature / °C**	**Duration / s** [1]
Denaturation	95	60
Annealing	50	60
Extension	72	60

Total *30* cycles

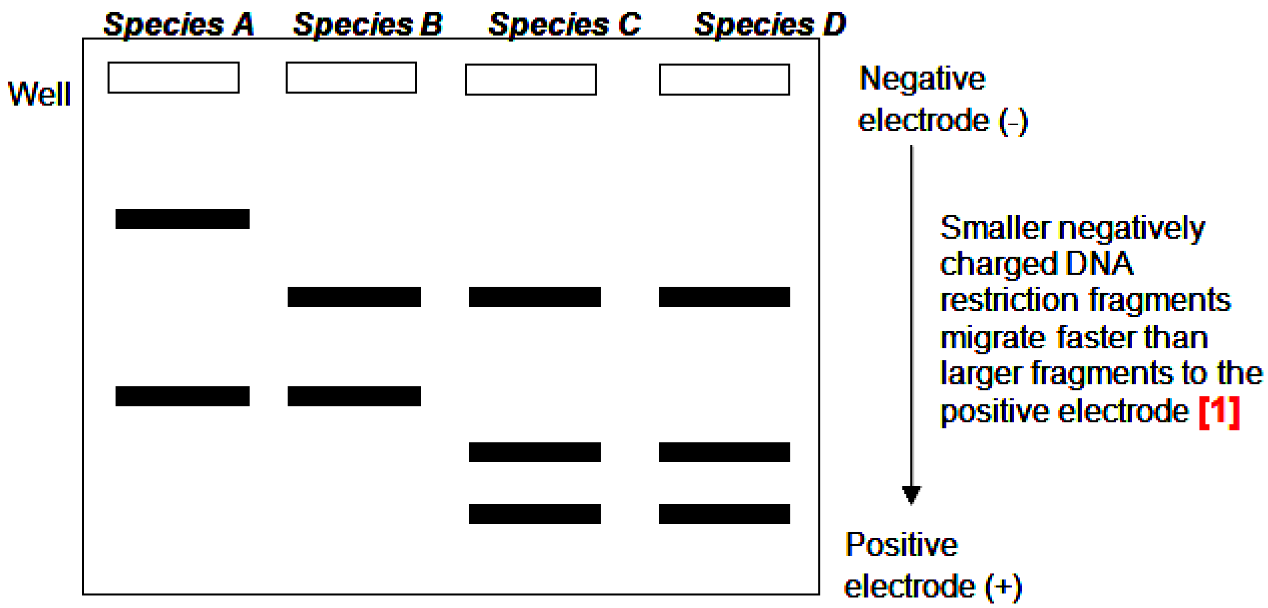
Reagents per PCR tube	**Volume / µl** [1]
DNA Taq polymerase mix including free DNA nucleotides	20
Forward DNA primer (1µM)	5
Backward DNA primer (1µM)	5
Buffer	5
DNA template (supernatant)	5

Note: Do not forget to include DNA template (i.e. DNA in the supernatant after DNA extraction) in the PCR tube / mixture

Digestion of extracted DNA *[note: after PCR]*
6) Digest the **10µl** amplified DNA with **1µl HindIII** in a **37°C** incubator for **1h**. [1]

Gel electrophoresis
7) Prepare 1% agarose gel. Pour the molten gel into gel casting tray. Insert the comb and let the gel set for 30 minutes.
8) Add **1µl loading dye** to **10µl of digested DNA**. Dispense the DNA samples into the wells of the gel. [1] *[Note: Add loading dye to DNA before loading into the well of agarose gel]*

9) Run at 100V for 1h.
10) Soak the gel in a solution of **ethidium bromide** and view the gel under **UV light** to compare the band patterns of the different species. [1]



Hypothetical results – drawn to depict correct evolutionary relationship or clearly described **[1]**

Safety issue **[1]**

Ethidium bromide (EtBr) is a carcinogen which is **harmful** to the body. **Avoid direct skin contact** with the chemical by **wearing gloves** while handling the EtBr-stained gel.

Free-response question

5 (a) Outline the procedure for cloning the human insulin gene using *E.coli* cells to produce human insulin for diabetic patients. Your answer should include how the eukaryotic gene is cloned to avoid the problems associated with introns. [10]

- Obtain the **mRNA of the insulin gene from** (islets of Langerhans of) **pancreas;**
- and **synthesize the cDNA** by **reverse transcription**;
- **Convert** the **single stranded cDNA to double stranded DNA** by first removing the mRNA using ribonuclease; and then synthesising the complementary strand using DNA polymerase;
- **Add additional guanine nucleotides;** to **create artificial sticky ends** for the insulin DNA using **terminal transferase**;
- **Cut plasmid** vector with **restriction enzyme** to **create blunt ends;** and **add additional cytosine nucleotides** to **create artificial sticky ends** for the plasmid;
- **Mix the cut plasmid and human cDNA;** to allow **hydrogen bonds between complementary bases** of the artificial sticky ends;
- **Create recombinant DNA** by addition of **DNA ligase** to link the nucleotide backbone;
- Introduce the **recombinant plasmid into host *E.coli* cells** by **transformation**;
- Reference to use of **calcium chloride and heat shock** method; to **increase uptake** of recombinant plasmid by competent *E.coli* cells;
- **Select for transformed *E.coli* colonies** that acquire the recombinant plasmid using **selective markers present on the plasmid;** / **replica plating** or **blue-white screening;** on appropriate **selection medium;** (i.e. antibiotic agar plates or agar plate containing X-gal and antibiotic respectively).
- **Description of identification** of host with recombinant plasmid;;
- **Culture** the transformed bacterial colonies in large quantities **in a suitable medium**;
- **Extract, purified and packaged the insulin** secreted for use in patient with diabetes;
- AVP, e.g. extraction of mRNA using matrix column containing oligonucleotides of thymine, advantage of using mRNA, expression of insulin gene in the presence of lactose, identify transformed bacterial colonies with the insulin gene by nucleic acid hybridization using radioactive gene probe and autoradiography, etc.

(b) Describe the advantages and limitations of PCR. [6]

Advantages	**Limitations**
Exponential amplification of DNA; Specific DNA sequences is being	Limited size range (0.1 to 5 kb); Entire genome cannot be copied;

synthesized/copied;	
Speed and ease of use / Can be performed in relatively short time, within a few hours	Requires DNA primers to be synthetically constructed; So only known sequences can be amplified / at least primer sequence has to be known;
High sensitivity; capable of amplifying minute amounts of target DNA; Robustness; dependability of the PCR process to continue operating well even though conditions are constantly changing;	Taq polymerase – no proof reading ability; cause inaccurate replication / mutations; Contamination; May amplify non-target sequences of DNA as DNA primers may be complementary to more than one portion of the DNA molecule
In vitro process (i.e. outside the cell); Easy to extract amplified DNA;	High temp (95^{o}C) is being used to denature (break hydrogen bonds) and separate the DNA into single strands; Repeated cycles of high heat treatment may damage DNA; Some DNA may fail to denature thus reducing the number of templates

(c) Discuss the ethical and social considerations of gene therapy [4]

<u>Ethical concerns</u>

- Tampering with human genes may lead to the practice of eugenics; the deliberate effort to control the genetic makeup of human populations;

- Manipulating organisms via transferring genes across natural boundaries; is seen as usurping the prerogative / right of God;

- Ethical concern due to risk of causing other diseases;
    - the use of retroviruses may cause mutation of other essential genes giving rise to other diseases such as cancer OR
    - Insertion of therapeutic DNA may activate oncogenes or disrupt tumour suppressor genes OR
    - Cells that are being modified may contain viruses that can help the vectors to spread to other cells OR
    - Germ-line gene therapy experiments would also involve too much scientific uncertainty and clinical risks / long term effects germ-line therapy are unknown;

- Risk of tampering with evolution of the human species with germ line therapy;
    - Genetic variety is a necessary ingredient for the survival of the species as environmental conditions changes with time; OR
    - Genes that cause harm in some environment may be of evolutionary significance in other environment; (e.g. sickle cell anaemia in malarial regions);
    - Making genetic changes could be detrimental to our species to adapt to changes in the future;

- Germ line therapy would allow the inserted gene to be passed on to future generations; this will create generations of un-consenting research subjects / violating their rights;

- Abuse of gene therapy for non-medical purposes / cosmetic reasons rather than for treating diseases; e.g. designer babies / people wanting to use gene therapy to enhance their basic human traits such as height, intelligence, or athletic ability.

- It is difficult to decide which traits are normal and which constitute a disability or disorder; no one has the right to decide too.

- It is difficult to determine the appropriate time in which gene therapy should be used → Babies / children, the critically ill.

- There may be issues on who should pay for gene therapy → Insurers / taxpayers / government.

- It is difficult to decide who should have access to gene therapy → Only those who can afford it.

Social concerns

- The high costs of gene therapy make it available only to the rich and powerful;

- The widespread use of gene therapy may cause society to be less accepting of people who are different / exacerbate problems of social discrimination;

[Total: 20]